

Module 2: HR Management

1. Overview

The HR Management module is the central repository for all employee data, with a critical focus on managing the skills, training, and certifications required by the Saudi Electricity Company (SEC). This module is not a standalone function; it is deeply integrated into the project execution workflow to ensure that only qualified and certified personnel are assigned to specific tasks. The primary goal is to maintain compliance, improve safety, and streamline team assignments.

2. Core Features

2.1. Employee Profiles

This is the foundation of the HR module. Each employee will have a comprehensive profile containing:

- **Personal Information:** Name, National ID/Iqama, Contact Information, etc.
- **Job Information:** Job Title, Department, Team Assignment, Employment Status.
- **Document Management:** A place to upload and store important documents like contracts, ID copies, and passport copies.
- **Certification Status:** A dedicated section to track all SEC-mandated courses and certifications.

2.2. Training & Certification Management

This is the most critical feature of the HR module.

- **Course Catalog:** A master list of all possible SEC courses and certifications (e.g., "Work At Height," "Low Voltage Permit to Work," "Excavation Safety"). Each course will have details like its validity period and the roles it is required for.
- **Employee Certification Tracking:** Each employee's profile will have a list of the courses they have taken. For each course, the system will track:
 - Course Name
 - Date of Completion
 - Expiry Date
 - Certificate Scan (PDF/Image)
 - Status (e.g., Active, Expired, Expiring Soon)
- **Automated Alerts:** The system will automatically send notifications to HR managers and employees when a certification is about to expire (e.g., 30, 60, 90 days out).

2.3. Team Management

- The system will allow for the creation of predefined teams (e.g., "Excavation Team A," "Electrical Team B," "Civil Team A").
- HR managers or department heads can assign employees to these teams.
- When a Project Manager assigns a team to a Work Order (e.g., in the Excavation tab), the system will automatically check if all members of that team have the required, active certifications for that type of work. If a team member has an expired or missing certification, the system will flag it and prevent the assignment until the issue is resolved.

2.4. Roles & Permissions

- The HR module will define user roles (e.g., HR Manager, Project Manager, Team Leader, Employee).
 - Permissions will be granular. For example, only an HR Manager can add or edit employee profiles, while a Project Manager can only view team certifications.
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3. Integration Points

- **Work Order Management (Module 1):**
 - **Team Assignment:** When a PM selects a team in the Excavation, Electrical, or Civil tabs, the system queries the HR module to validate the certification status of every team member against the requirements of that work type.
 - **Form Auto-Filling:** Employee names and ID numbers for forms like "Contractor Team Work" will be pulled directly from the HR module.
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4. Database Schema Outline

To support this functionality, the following database tables (or similar) would be required:

- employees: Stores all personal and job information for each employee.
- courses: The master list of all SEC certifications.
- employee certifications: A linking table that connects employees to the courses they have completed, including completion and expiry dates.
- teams: Defines the various work teams.
- team members: A linking table that assigns employees to teams.
- course requirements: A table that maps specific job roles or work types (e.g., "Excavation") to the courses required for them.